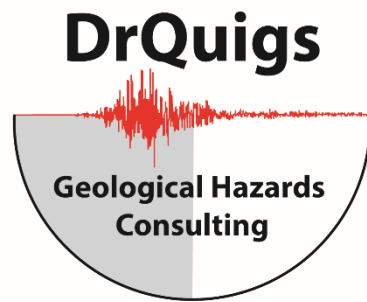




MEMO TO SA WATER REGARDING SEISMIC HAZARD ANALYSIS AT MOUNT BOLD

TO: Valiyff, Najib <najib.valiyff@sawater.com.au>; Binti Musa, Erina <erina.bintimusa@sawater.com.au>; Walshe, Ciaran <ciaran.walshe@sawater.com.au>

FROM: Dr. Mark Quigley (Director DrQuigs Geological Hazards Consulting Pty Ltd, Associate Professor Earthquake Science at the University of Melbourne)

DATE: 6 August 2025

OVERVIEW:

In a brief meeting held on 4 August 2025, SA Water requested Dr. Quigley provide the following:

- (i) Slip rate advice and parameters for the Willunga Fault for Probabilistic Fault Displacement Hazard Analysis (PFDHA) and Probabilistic Seismic Hazard Analysis (PSHA)
- (ii) Summary statement regarding the 3D geological model for the Willunga Fault
- (iii) Summary statement regarding the Ochre Cove – Clarendon Fault
- (iv) Summary statement regarding the Meadows Fault

These items are provided below.

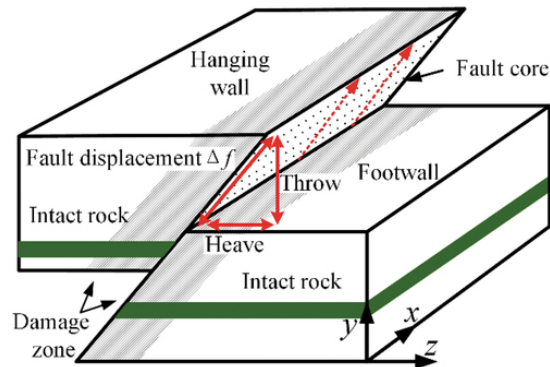
Slip rate advice and parameters for the Willunga Fault for Probabilistic Fault Displacement Hazard Analysis (PFDHA) and Probabilistic Seismic Hazard Analysis (PSHA)

Fault slip rate is an important parameter for both probabilistic seismic hazard analysis (PSHA) and probabilistic fault displacement hazard analysis (PFDHA).

Fault slip rate (F_{SR}) is computed as:

$$F_{SR} = \text{Fault displacement} / \text{Time}$$

Fault displacement (Δf in image below) is defined by the amount of displacement of a geological horizon associated with slip on a defined fault. In the Australian context it is common to measure the vertical displacement (i.e., 'Throw') and use inferences or direct observations of the fault dip to estimate Δf using trigonometry. In some instances, it may be possible to determine Δf directly (e.g., in natural cross-sectional exposures or trenches). Epistemic uncertainties on fault geometry may be substantial; commonly this is addressed using probabilistic and logistic approaches that incorporate diverse estimates for fault dip (and potentially curvature and fault width). Estimates of displacement across discrete structures may underestimate total displacements across broader fault zones.



Time refers to the time interval over which the computed Δf accumulated. In many cases this 'Time' estimate is computed from the inferred age of a faulted geological unit or surface. This parameter is typically highly uncertain and poorly constrained, resulting in large epistemic uncertainties in computed fault slip rates that can be characterised through probabilistic and logistic approaches that incorporate diverse estimates for *Time*. A related source of major uncertainty is the tendency for Australian faults to undergo earthquake quiescence and clustering episodes with lower or higher F_{SR} relative to the longer-term F_{SR} average. This requires expert judgement and is an epistemic uncertainty in terms of whether future earthquakes will adhere to the spatial-temporal patterns of prior ones encoded in the geological record.

For the purposes of PSHA, it is typical to use a spatially and temporally 'averaged' F_{SR} for the entire mapped fault. Because displacements and time intervals of activity may vary significantly along faults, the average F_{SR} may be significantly different from the F_{SR} at a given location.

For the purposes of PFDHA, it is typical to use the F_{SR} for a specific location.

An extensive volume of work has been conducted at Mount Bold and elsewhere along the Willunga Fault to establish the average F_{SR} for PSHA (Appendix A) and local F_{SR} for PFDHA (see Appendix B).

DrQuigs GHC Pty Ltd has reviewed this work, and the parameters are recommended:

Average F_{SR} for PSHA (Willunga Fault):

MEDIAN (50th percentile) F_{SR} : 50 m / Myr (0.05 mm a⁻¹)

85th fractile F_{SR} : 60 m / Myr (0.06 mm a⁻¹)

95th fractile F_{SR} : 90 m / Myr (0.09 mm a⁻¹)

Local F_{SR} for PFDHA (Willunga Fault):

MEDIAN F_{SR} : 18 m / Myr (0.018 mm a⁻¹)

85th fractile F_{SR} : 44 m / Myr (0.044 mm a⁻¹)

95th fractile F_{SR} : 70 m / Myr (0.70 mm a⁻¹)

Summary statement regarding the 3D geological model for the Willunga Fault

A 3D geological model for the Willunga Fault at Mount Bold was produced by GHD in 2017:

Macklin, S. R., Terzic, Z., Barter, J. F., Buchanan, P., & Quigley, M. (2019). A multi-disciplinary approach to active fault rupture risk characterization: 3D geological modelling of the Willunga fault, Mt Bold Dam, South Australia. In Sustainable and Safe Dams Around the World/Un monde de barrages durables et sécuritaires (pp. 2984-2997). CRC Press.

Dr. Quigs GHC does not have access to this article or model. SA Water is encouraged to seek access to this article and model if they do not have it.

An updated geological model for the Willunga Fault at Mount Bold is being produced by SMEC. Dr Quigs GHC is contributing to this model development.

A 3D model for the entire Willunga Fault is currently being developed by Dr. Quigley and his research team (including Geoscience Australia) through a project funded by SAFECOM (South Australian Fire and Emergency Services Commission) and NEMA (National Emergency Management Agency). Geoscience Australia is a partner in this collaboration. When this model is completed (anticipated early 2026) it will be provided directly to SA Water and any other interested parties.

The 3D model of Mt Bold is required for PFDHA and other engineering applications. The 3D model of the entire Willunga Fault can enhance PSHA but is not required for PSHA.

Summary statement regarding the Ochre Cove – Clarendon Fault

In NSHA 23, the Ochre Cove – Clarendon Fault is assigned a **slip rate of 42 m / Myr (0.042 mm a⁻¹)**. For details on the computation of this slip rate and estimates of other relevant parameters for PSHA please see <https://neotectonics.ga.gov.au/?execution=e1s1>

This slip rate is poorly constrained. LiDAR is available for this fault and more reliable slip rate calculations could be performed as part of a portfolio of future work involving mapping and analysis.

In the absence of any future work, the NSHA 23 value is recommended.

Summary statement regarding the Meadows Fault

In NSHA 23, the Meadows Fault is assigned a **slip rate of 13 m / Myr (0.013 mm a⁻¹)**. For details on the computation of this slip rate and estimates of other relevant parameters for PSHA please see <https://neotectonics.ga.gov.au/?execution=e1s1>

This slip rate is poorly constrained. LiDAR is available for this fault and more reliable slip rate calculations could be performed as part of a portfolio of future work involving mapping and analysis.

In the absence of any future work, the NSHA 23 value is recommended.

For more details or queries regarding anything in this report, please contact Mark Quigley:

Mark.quigley@unimelb.edu.au

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APPENDIX A

From Wilson (2008):

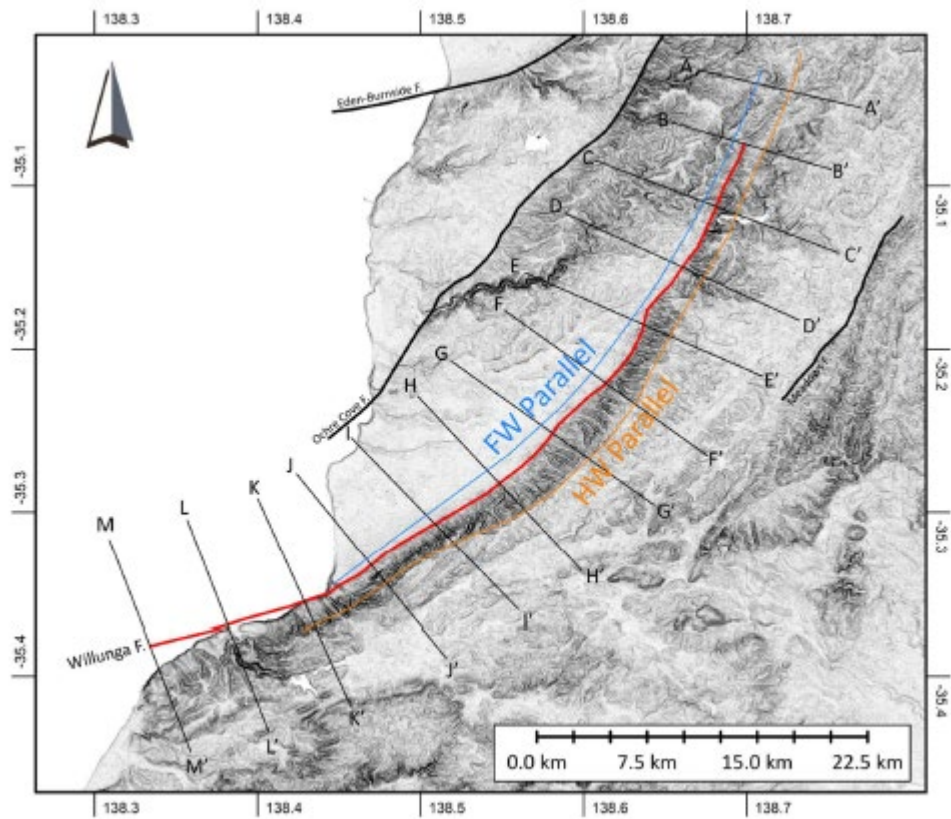
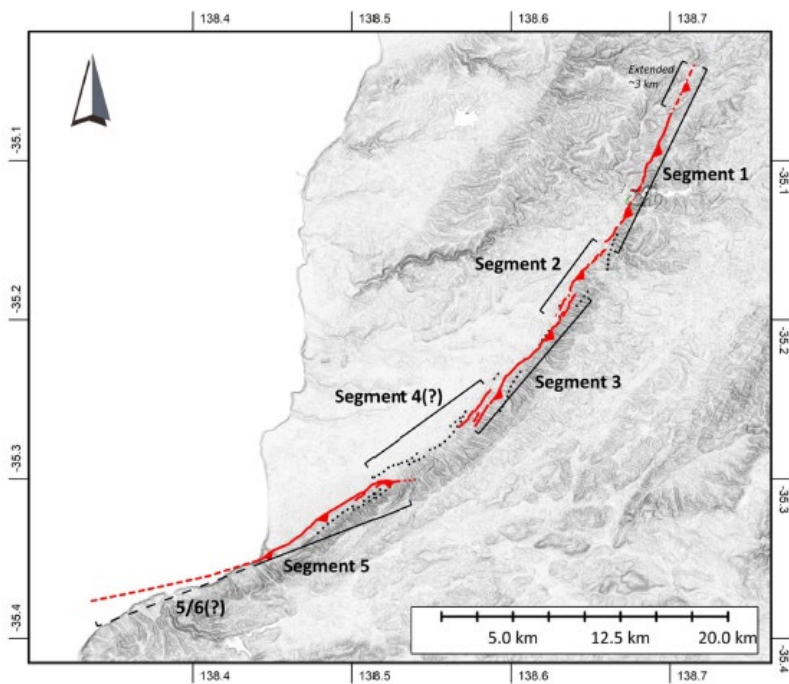


Figure 2.21. Parallel (FW and HW) and perpendicular (A-M) profiles used to analyse elevation and displacement across the entire Willunga Fault.

From Wilson (2008):



From Wilson (2008):

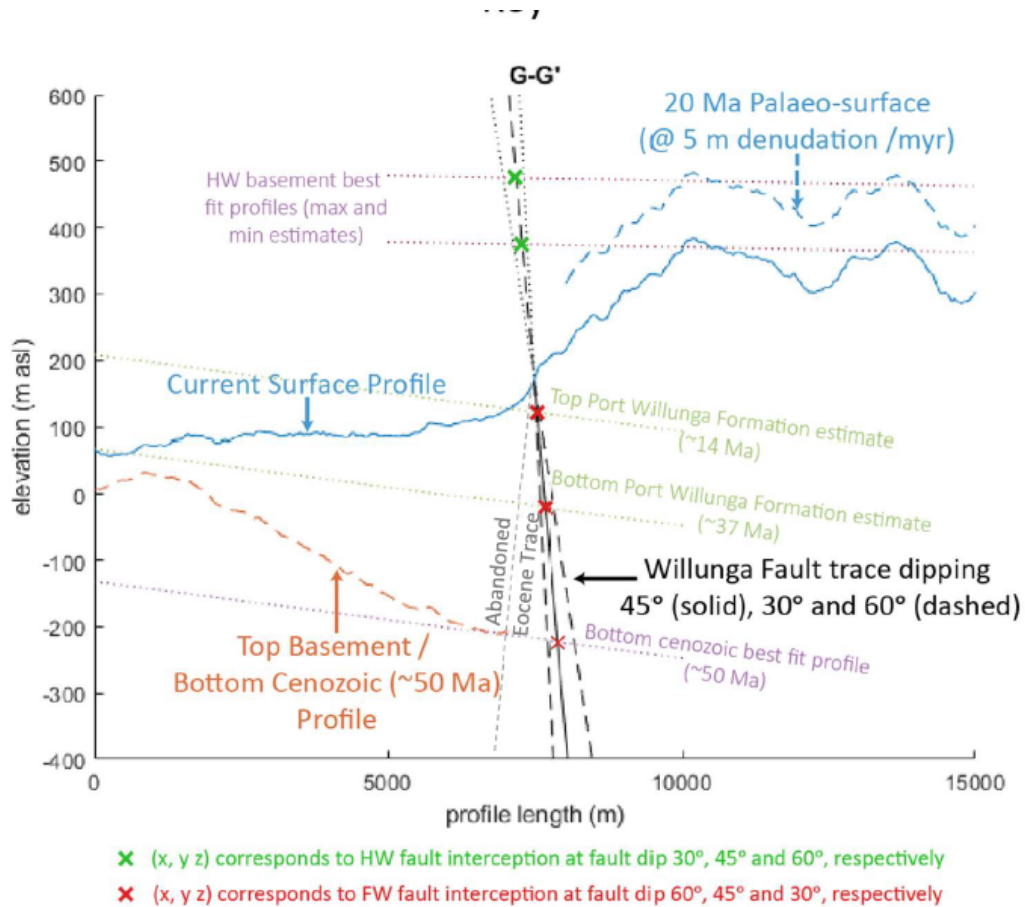


Figure 2.22. Example of Fault perpendicular profile G and key to how each profile (A-M) was dissected and analysed. Depth to certain stratigraphies can be seen in the FW (left of the Fault) and example of palaeo-surface at 20 Ma can be seen on the HW (right of the Fault.) Multiple Fault dips have been shown too.

From Wilson (2008):

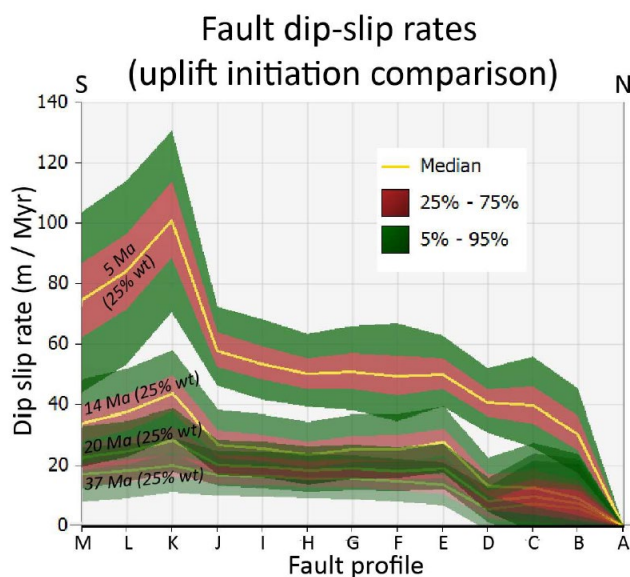


Figure 6.1. Comparison of 5, 14, 20, and 37 Ma calculated dip-slip rates (m / Myr) of the Willunga Fault (A-M)

From Wilson (2008):

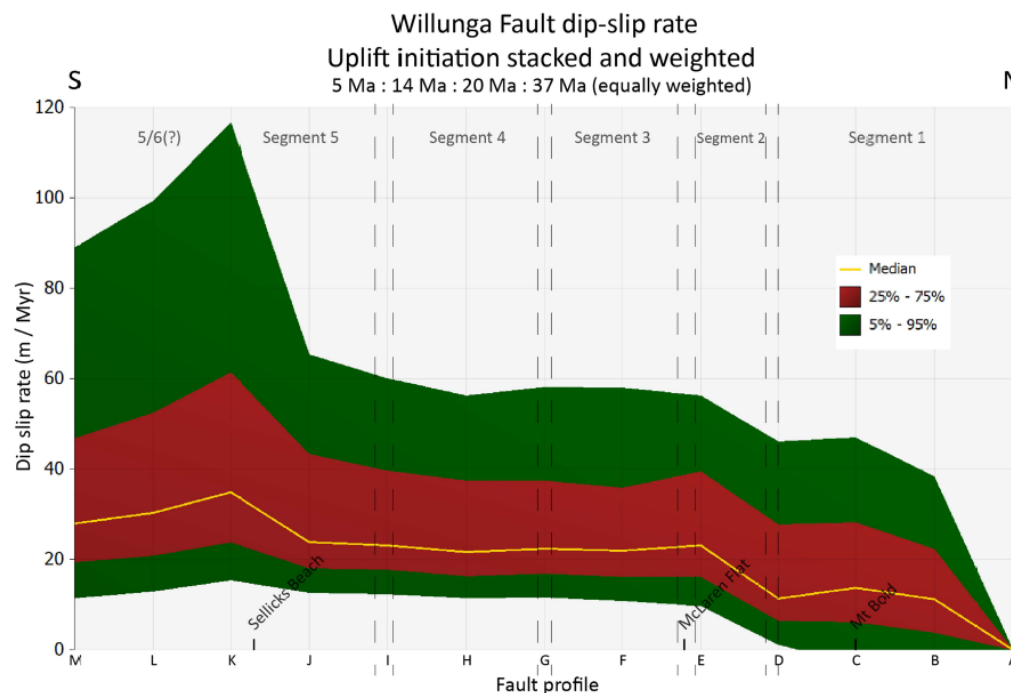


Figure 6.2. Stacked and weighted dip-slip rate (m / Myr) of the Willunga Fault (A-M). Geometrical segment extents and study site locations are overlain.

From Clark et al. 2024 (for approximate area of Profile J from Wilson 2018)

“...southern Willunga Fault has constrained the timing of six surface-rupturing earthquake events in the last ca. 150 ka, associated with a potential slip rate in the order of ~65-70 m/Myr”

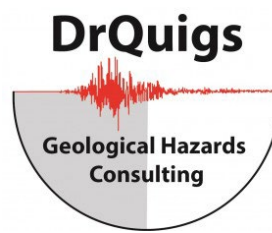
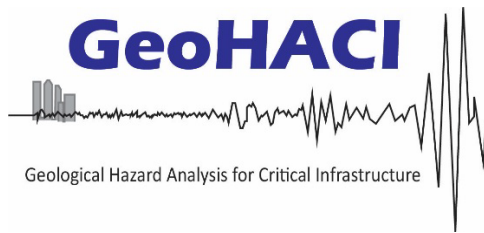
From Australian Neotectonics Database:

Slip Rate Remarks: Sandiford (2003) suggest a time averaged slip rate of 0.05-0.07 mm/a across the fault based upon the raised subtidal deposits of last interglacial age reported from Normanville (see Bourman et al. 1999). This estimate is independently corroborated reported elevation differences of ~130 m in the Lower Pleistocene (ca 1.7 Ma) Burnham Limestone along the eastern margin of the St Vincents Basin (Belperio 1995; Bourman et al. 1998). However, uplifted last interglacial benches and trenches excavated across the fault trace south of Cactus Canyon (Sellicks Beach) indicate only ~2m of uplift at this location, suggesting that the slip rate could be as low as 0.02 mm/a. The differential elevation of the Port Willunga Formation between the Myponga Basin and the St Vincents Basin suggests a post-Early Miocene vertical displacement of ~240 m (Tokarev et al 1999), which suggests a long term slip rate averaged over most of the Neogene (<15 Ma) of ~0.02 mm/a.

From NSHA 23:

Slip Rate for the Willunga Fault = 0.092 mm a^{-1}

APPENDIX B



WILLUNGA FAULT SOURCE PARAMETERS FOR UTILITY IN PFDHA: FAULT DIP AND SLIP RATE AT MOUNT BOLD

Prepared for:

GHD, SA Water and AECOM

Prepared by:

Mark Quigley
Associate Professor of Earthquake Science
University of Melbourne
Director of GeoHACI and DrQuigs GHC
T: 0415900152

Submitted on:

14 June 2019

EXECUTIVE SUMMARY

At the request of Dr. Paul Somerville (AECOM) (email.dated.1June.8676).and with the consent of GHD and SA Water, I herein provide expert advice on the slip rate of the Willunga Fault at Mount Bold, for utility in Probabilistic Fault Displacement Hazard Analysis (PFDHA).

Fault slip rates are determined from estimates of (i) vertical offsets of correlative geologic units and/or surfaces, (ii) fault dip, (iii) slip vector, and (iv) time estimates over which finite offsets accumulated. These parameters are provided herein. The provision of uncertainties with these estimates enables direct implementation into logic trees and probabilistic analysis.

On the basis of this work, I propose that:

1) The **preferred central estimate (with 1 σ error) for the strike and dip angle of the Willunga Fault at Mount Bold is $020 \pm 9^\circ$ (strike) and $43 \pm 4^\circ$ (dip) to the east-southeast.** This estimate equally weights (i) the average of 10 measurements of the fault at depths of 0 to 200 m derived from 3-D LeapFrog™ modelling of best-fit fault geometry based on seismic reflection, seismic refraction, borehole data, and surface fault data inputs, and (ii) the average of two measurements of the scarp-producing principal late Quaternary fault slip zone as identified in two paleoseismic trenches, and (iii) the results of two-point solutions for two fault plane fitting solutions based on neotectonic fault traces north of the proposed auxiliary spillway. Based on this direct evidence, I propose the following **logic tree weights for dip angle in PFDHA: 35° (0.2); 45° (0.6), 55° (0.2).** The previous estimates by AECOM (2019) in their source recurrence model for PFDHA ($45^\circ = 0.2$; $60^\circ = 0.6$; $75^\circ = 0.2$) are not supported by any available evidence at this site and should be succeeded by the estimates provided herein.

2) The **preferred, central weighted estimate of the time-averaged slip-rate for the Willunga Fault at Mount Bold is 0.018 mm/yr (+.03, -0.003 mm/yr at 68% confidence).** This estimate is based on measurements of the vertical offset across correlative planation surfaces on opposite sides of the fault at Mount Bold and weighting of estimates of fault inception age (5, 14, 20, 37 Ma). This is consistent with the previous estimate of 0.027 ± 0.010 mm / yr reported by GHD (2017) for the Mount Bold site. It is lower than the value reported for the entire Willunga Fault by Leonard et al. (2014) however that data is assembled from fragmentary and highly uncertain data on distal parts of the fault that have no direct bearing on the Mount Bold site for the purposes of PFDHA. Based on this analysis, we recommend the following **logic tree weights for fault slip rate in PFDHA: 0.006 mm/yr (0.15); 0.009 mm/yr (0.25); 0.025 mm/yr (0.4) 0.038 mm/yr (0.2).** These weightings and values are distinct from (and succeed) those presented in AECOM (2019); their estimates are assigned based on Leonard et al. (2014) without justification or evidence for their utility.

	inception age (Ma)				Slip rate (mm/yr)				Dip deg		
Weight	0.75	0.15	0.07	0.03	0.15	0.25	0.4	0.2	0.2	0.6	0.2
Willunga Fault	5	14	20	37	0.006	0.009	0.025	0.038	35	45	55

3) Recommendations to revise AECOMs (2019) weightings for seismogenic thickness and PFDHA curves are also provided.

1. Strike and dip angle of the Willunga Fault at Mount Bold

1.1. LeapFrogTM fault model inputs

LeapFrogTM software has been used by GHD (J. Barter) (Figure 1) to produce best-fit models for the Willunga Fault strands to (i) mapped surface locations of faults, (ii) locations of faults imaged in seismic refraction surveys, (iii) locations of faults imaged in seismic reflection surveys, and (iv) locations of faults detected in boreholes. The geophysical and structural data is provided in GHD (2017) and the models are available by contacting James.Barter@ghd.com. Ten measurements were obtained 0 to 200 m depth from the best-defined fault trace proximal to the saddle dam (Table 1). The average fault dip / dip direction is 47 / 115.

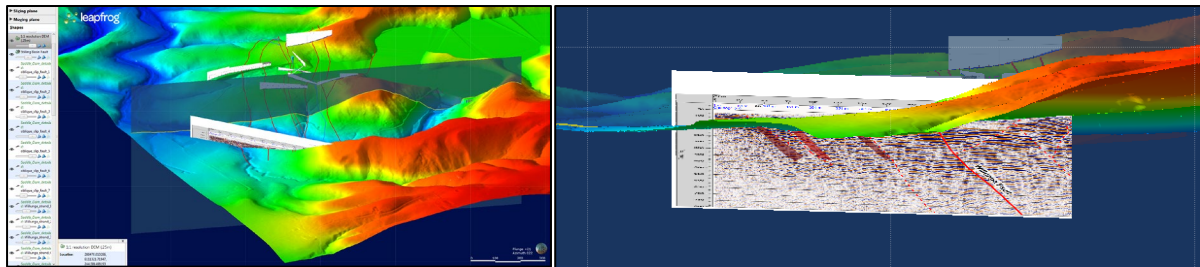


Figure 1: 3-d model showing locations of seismic lines used to constrain fault dip.

Fault dip [deg]	Dip dir [az]
47	131
47	123
47	113
48	115
46	106
42	105
49	119
48	109
46	115
47	114

Table 1: Measurements of Willunga Fault strand 1, from best-fit LeapFrogTM model. Average = 47 ° dip at 115° dip direction.

1.2. Fault dip in paleoseismic trenches

MT BOLD TRENCH 1 NORTH

GHD (2017) excavated paleoseismic trenches across the Willunga Fault trace at three locations (Figure 2). At Trench 3 North, red-weathered Quaternary colluvial breccia was found to be offset by 63±5 cm along the reverse fault, and this was interpreted to represent the discrete most

recent surface rupture event (MRE) displacement. The fault plane is manifested as a ~12cm thick crush zone at the base of the trench and strikes 030° and dips 40°E (Figure 3).

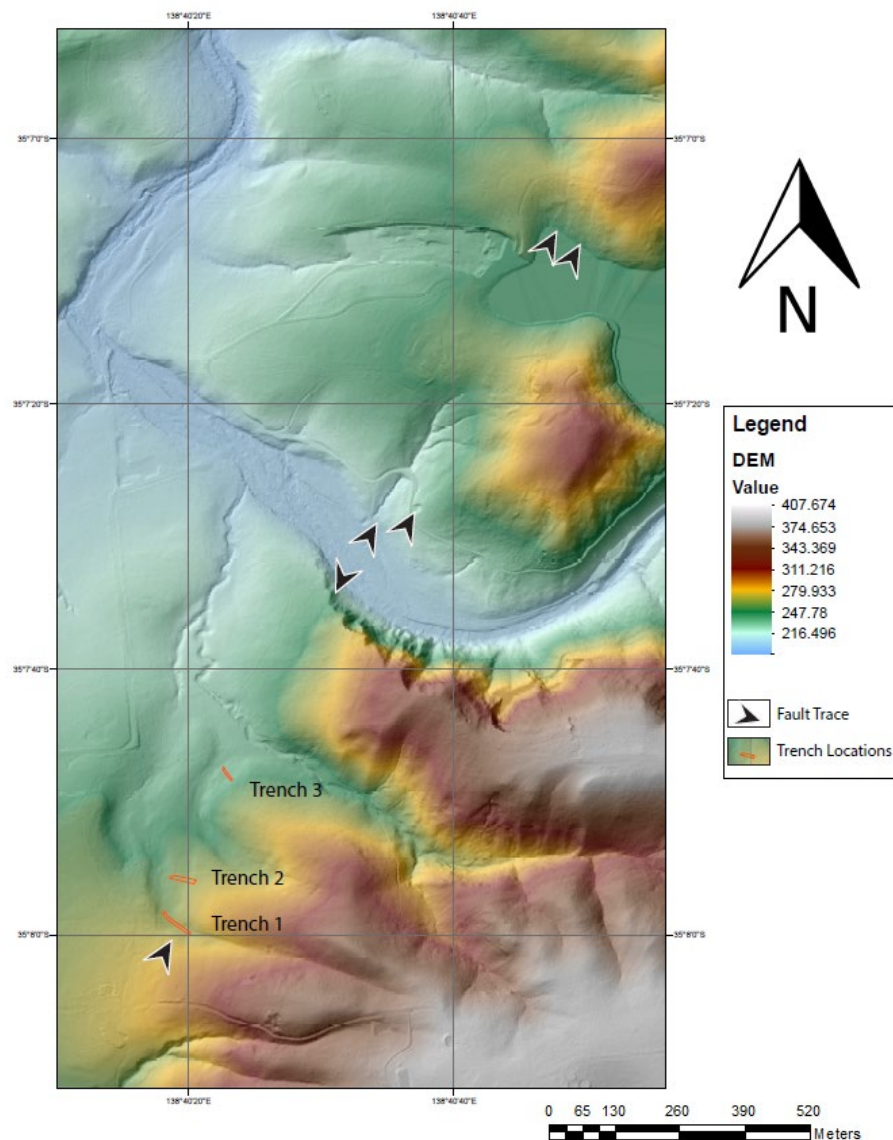


Figure 2: The location of the three trenches on the LiDAR DTM. The Quaternary Willunga Fault surface trace is denoted with black arrows; see GHD (2017) for more detailed fault maps.

MT BOLD TRENCH 2 CENTRAL

In Mount Bold Trench 2 Central, the Willunga Fault plane is directly observed as a reverse fault structure where weathered bedrock was thrust over Quaternary clay-rich sediments. The fault plane was measured, with a 40° dip towards 092° and is aligned with the scarp identified in the LiDAR imagery.



Figure 3: Field photograph from paleoseismic Trench 3 North showing fault zone (pink markers).

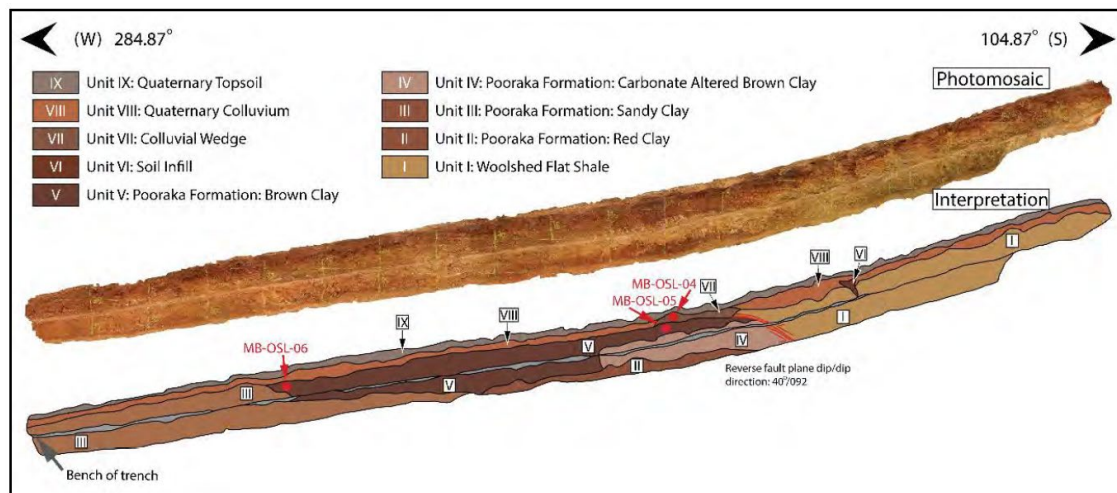


Figure 4: Photo and interpretation of the Mount Bold Central 2 Trench. The fault zone is marked in red, with fault plane measurements listed underneath (40° dip towards 092°). From Wilson (2018).

1.3. Fault dip from three-point solution from LiDAR data

Using the LiDAR data, I measured the height of surface traces at the base of the exposure (dam water level) and at the ridge crest for two prominent strands of the Willunga Fault northeast of the proposed saddle dam location (Figure 5). The horizontal, fault-normal separation between top and bottom exposures was used to calculate fault dip. The average fault dip of the two strands is 43° towards 110°.

1.4. Conclusion

Three independent, equally robust methods to determine the dip of the Willunga Fault indicate an average strike of $020 \pm 9^\circ$ and dip of $43 \pm 4^\circ$ (dip) to the east-southeast. To account for the observed range and epistemic uncertainties, I propose the following **logic tree weights for dip angle in PFDHA: 35° (0.2); 45° (0.6), 55° (0.2)**. The previous estimates by AECOM (2019) in their source recurrence model for PFDHA (45° = 0.2; 60° = 0.6; 75° = 0.2) are not supported by any available evidence at this site and should be succeeded by the estimates provided herein.

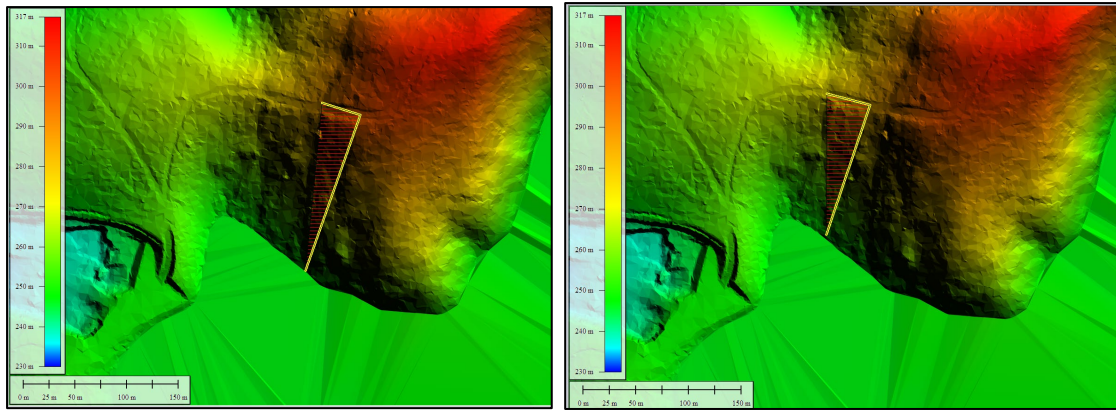


Figure 5: Quaternary surface rupture traces on the Willunga Fault (strike = 020°). Location of surface trace at base of exposure (southern end of yellow line), and horizontal separation to location of same trace on ridge crest (north western end of yellow line) was used with spot elevations to estimate fault dip. The eastern trace (left panel) dips 50° towards 110°; the western trace (right panel) dips 35° towards 110; the average fault dip is 43° towards 110°.

2. Slip rate of the Willunga Fault at Mount Bold

2.1. Preceding work

GHD (2017) and Terzic et al. (2017) obtained a Willunga fault slip rate at Mount Bold of 27 ± 10 m / Myr assuming a 5 to 10 Ma fault inception age and fault dip range of 40 to 60°.

Leonard et al. (2014) stated fault slip rates of (i) 60 m / Myr (since 0.12 Ma), (ii) 100 m / Myr (since 1.7 Ma), and (iii) 20 m / Myr (since 15 Ma). AECOM (2019) used the latter estimates in their analysis.

The (i) slip rate provided by Leonard et al. (2014) is estimated from a proposed “ 7 ± 1 m” of vertical offset of a last interglacial bench at Sellicks Beach, more than 35 km from Mount Bold on a different segment of the Willunga Fault (Wilson, 2019). The most robust study of the outcrop from which this offset is from May and Bourman (1984) who “identify the remnant of a former shore platform on the uplifted Willunga Fault Scarp at an elevation of 4–5 m at Sellicks Beach” (Preiss, 2019). This sub-horizontal surface was cut across the steeply dipping beds of the Port Willunga Formation, and beach gravels resting on this surface have been correlated with last interglacial sediments by amino acid racemisation dating (Murray-Wallace & Bourman, 2002). The equivalent (last interglacial) shoreline in the proximal, tectonically- stable Eyre Peninsula ranges from 2.1 ± 0.5 to 4 ± 0.5 m above present sea-level (Murray-Wallace et al., 2016). Thus, depending on which constraint is used, **the actual 0.12 Ma rate estimated from this location is 0 to 25 m / Myr**. The Leonard et al. (2014) rate used by AECOM is without evidence at this location. Furthermore, the Willunga Fault at this location was active during Oligo-Miocene deposition (Preiss, 2019), suggesting a protracted history that is different from the Willunga Fault at Mount Bold. This is particularly evident in the marked changes in the total offset across fault as measured from basement rocks (Figure 6) (Wilson, 2018).

The justification for the other rates from Leonard et al. (2014) used in the analysis by AECOM (2019) cannot be evaluated because no information is provided by either as to where or how these rates were estimated. This is particularly relevant in terms of the proposed assumption of a »major.change.in.the.local.stress.field.at.0.Ma” (Leonard et al., 2014; AECOM, 2019); if there is

any evidence for this it would instructive to declare it, however to the best of my knowledge no such evidence exists.

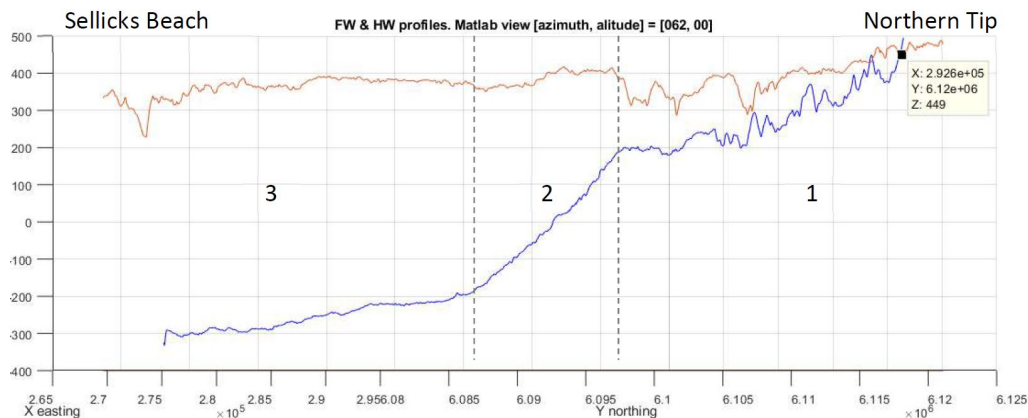


Figure 6: Topographic profiles showing the dramatic along-fault variation in depth-to-basement rocks (blue line) on the footwall of the Willunga Fault from the Mount Bold area (proximal to northern tip) to Sellicks Beach. The brown line indicates elevation of basement rocks on the hangingwall. The data indicate that the Willunga Fault comprises large along-fault differences in slip rate and/or fault inception age. This is discussed in greater detail in Quigley (2019). Data from the southern Willunga Fault are not easily interpreted in terms of relevance to the northern Willunga Fault.

2.2. Slip rate of the Willunga Fault at Mount Bold

We measured minimum, mean, and maximum elevations from a fault-perpendicular profile across the Mount Bold area (Figure 7).

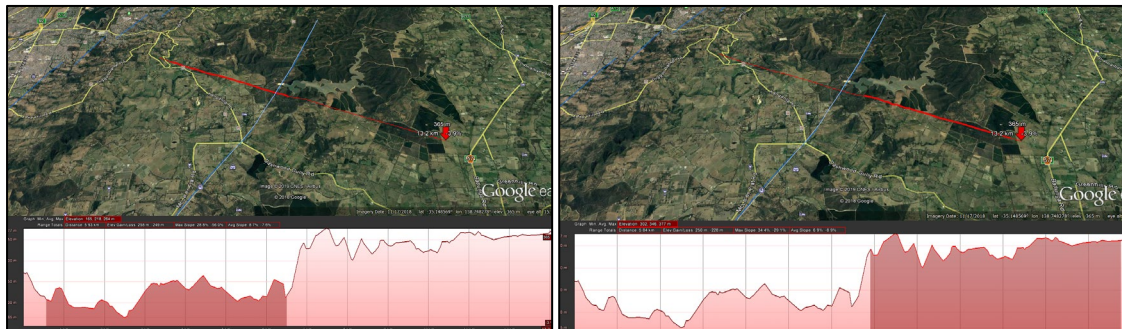


Figure 7: Topographic profiles across the Willunga Fault. Profile C intersects the Mount Bold Dam and Reservoir.

I calculate all fault parameters using the @Risk Excel add-in by Palisade Decision Tools to perform a 10,000 iteration Monte Carlo simulation based on field and geophysical measurements. I calculate the throw (D_v - vertical component of displacement) on the fault by differencing the elevations of the footwall and hanging wall surfaces, to which we assign the PERT distributions shown in Figure 8 (footwall) and Figure 9 (hanging wall).

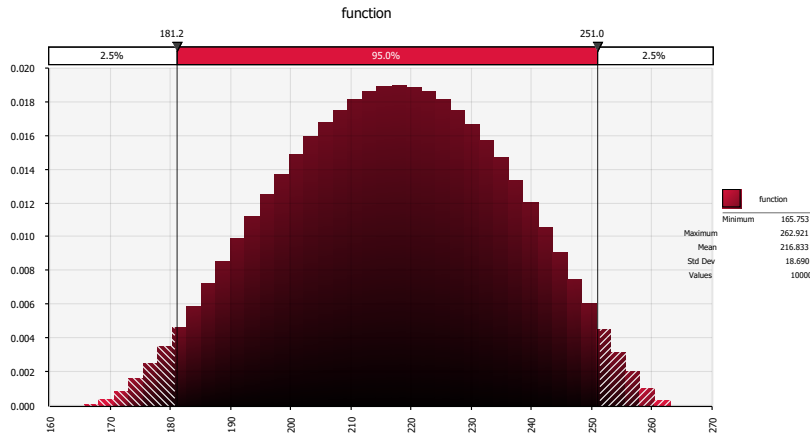


Figure 8: Footwall elevation range (Pert).

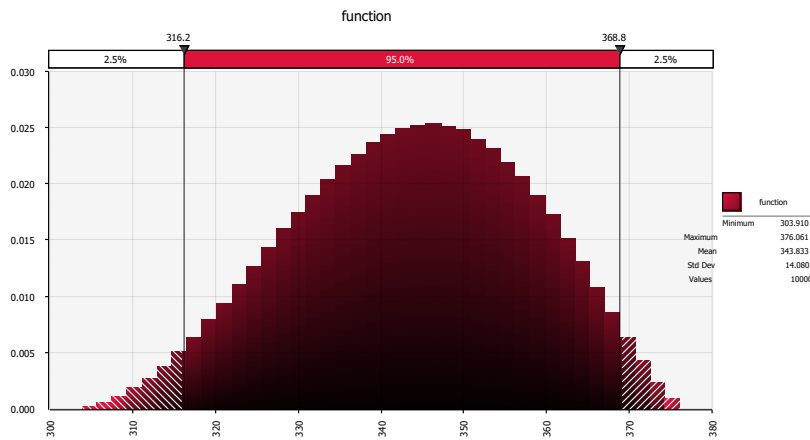


Figure 9: Hanging wall elevation range (Pert).

Net slip is then given by the equation:

$$D = \frac{D_v}{\sin \alpha}$$

Where D is the net slip, D_v is the throw estimated above and η is the dip of the fault. Our measurements (Table 1) establish the fault dip as $\approx 45^\circ$. We conservatively use a range of fault dips between 30° and 60° with a peak probability of 45° (Figure 10) to estimate that net slip $D = 182 \pm 39$ m (68%) (Figure 11).

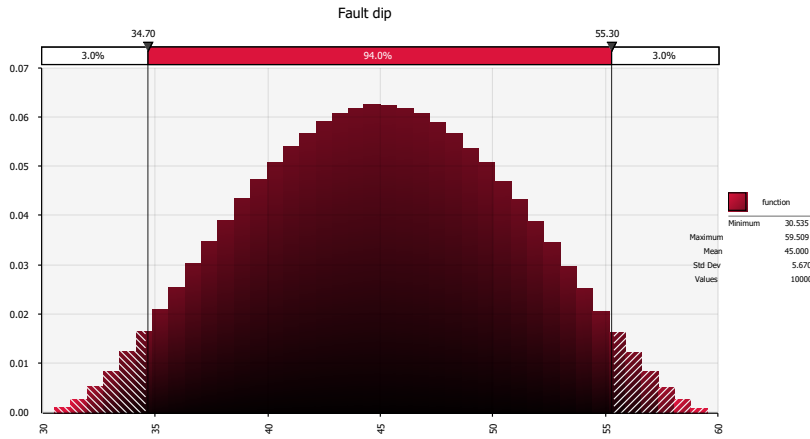


Figure 10: Fault dip distribution, centred on a dip of 45 ° (Pert)

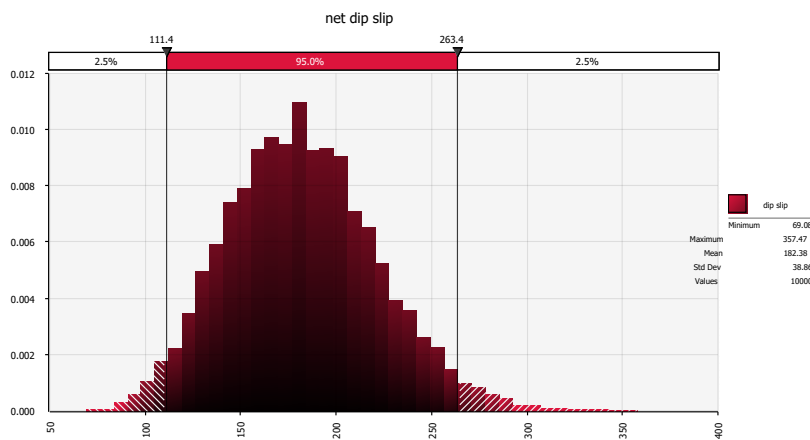


Figure 11: Monte Carlo modelled magnitude of net slip.

We calculate the slip rate (SR) from displacement and time (t):

$$SR = \frac{D}{t}$$

We use a log-normal distribution for the age of inception of slip in the neotectonic regime, to reflect our confidence that neotectonic slip initiated during (and probably early in) the Pliocene. The age distribution is truncated at 2.5 and 45 Ma and has a peak probability of inception at 5 Ma (Figure 11). Monte Carlo simulations of slip rate using the net slip in Figure 11 and the inception age distribution in Figure 12 yield a slip rate of 14-44 m/Myr, with a peak probability at a slip rate of 18 m/Myr (Figure 13).

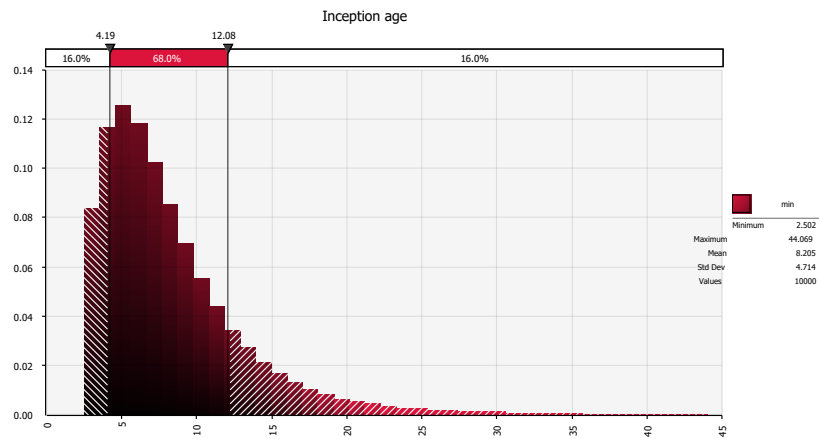


Figure 12: PDF for age of inception of neotectonic slip.

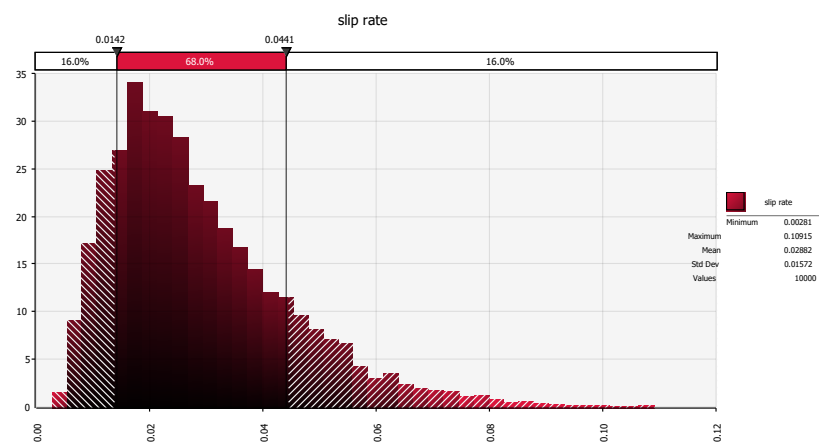


Figure 13: Monte Carlo modelled slip rates for the Willunga Fault at Mt Bold.

Based on the distributions estimated and calculated here, I recommend logic tree weightings for fault parameters as indicated in Table 2.

	Dip deg			inception age (Ma)				Slip rate (mm/yr)			
Weight	0.2	0.6	0.2	0.75	0.15	0.07	0.03	0.15	0.25	0.4	0.2
Value	35	45	55	5	14	20	37	0.006	0.009	0.025	0.038

Table 2: Recommended weighting for Willunga fault dip and slip rate at Mount Bold.

2.3. Conclusion

An updated probabilistic analysis of the Willunga Fault slip rate at Mount Bold reveals a slip rate of 0.014-0.044 m/Myr, with a most probable slip rate of 0.018 m/Myr.

I recommend logic tree weights for fault slip rate in PFDHA: 0.025 mm/yr (0.4); 0.009 mm/yr (0.25); 0.006 mm/yr (0.15); 0.038 (0.2). The method and slip rates used in AECOM (2019) are without an evidence base and thus the rates presented herein succeed the AECOM inputs.

3. Other comments on the AECOM (2019) PFDHA report

3.1. Seismogenic thickness

AECOM (2019) use weighting for seismogenic thickness of 0.1 (12 km), 0.8 (15 km), 0.1 (20 km) but did not provide detail as to the evidence base for selecting these values. As there has not been a large historical surface rupturing earthquake in the Mount Lofty - Flinders Ranges seismic zone, which includes the Mount Bold area, it is challenging to evaluate this aspect. All historical surface-rupturing earthquakes in Australia and worldwide in non-extended (cratonic) stable continental region crust have hypocentral depths < 5 km (Figure 14; from Clark et al., 2013) however the extent to which these offer appropriate analogous to the extended crustal terrain of the Mount Lofty - Flinders Ranges is uncertain.

In my expert opinion, the most appropriate approach is to weight the probability of a shallower (≤ 12 km) depth more highly to account for the empirical record of historical earthquakes (Figure 14), the earthquake depth distribution outside the Flinders Ranges (Figure 15B), and crustal strength profiles (Figure 15C), and similarly increase the weight of a deeper earthquake source for the same reasons (Figure 15C, 15D). This approach better accounts for the large epistemic uncertainty in considering hypocentral depths for future ground-rupturing earthquakes on the Willunga Fault. If the weightings are best presented using AECOMs (2019) binned depth estimates, **I recommend 0.4 (12 km), 0.4 (15 km), 0.2 (20 km).**

Surface-Rupture Length (L_{sr}), Seismic Moment (M_0), and Moment Magnitude (M_w) Data for Historic Surface-Rupturing Earthquakes in Nonextended SCR Crust Worldwide

Date (yyyy/mm/dd)/Event	Hypocentral Depth (km)	L_{sr} (km)*	L_{sr} Range*		M_0 (N·m)	M_0 Range		M_w	Hypocentral Depth/Seismic Moment Reference
			Min	Max		Min	Max		
1968/10/14 Meckering	3.0	37	36	38	8.32×10^{18}	6.10×10^{18}	1.04×10^{19}	6.58	Vogfjörd and Langston (1987)
1970/03/10 Calingiri	1.0	3.3	3.3	3.3	1.72×10^{17}	1.57×10^{17}	1.92×10^{17}	5.46 [†]	Denham <i>et al.</i> (1980)/Johnston (1994)
1979/06/02 Cadoux	2.7	14	14	16	1.73×10^{18}	1.22×10^{18}	2.44×10^{18}	6.13	Boatwright and Choy (1992)/GCMT
1986/03/30 Marryat Creek	4.5	13	13	13	4.57×10^{17}	3.24×10^{17}	6.46×10^{17}	5.74	Boatwright and Choy (1992)/GCMT
1988/01/22 Tennant Creek (Σ) [‡]	3.6, 3.5, 4.5	36 [§]	32	40	1.55×10^{19}	1.10×10^{19}	2.20×10^{19}	6.76	Boatwright and Choy (1992)/GCMT
1989/12/25 Ungava, Canada	3.0	8.5	4.5	13	1.12×10^{18}	7.92×10^{17}	1.58×10^{18}	6.00	Bent (1994)/GCMT
1993/09/29 KillariLatur, India	2.6	3	1.0	3.0	2.21×10^{18}	1.56×10^{18}	3.12×10^{18}	6.20	Seeber <i>et al.</i> , 1996/GCMT
2007/10/09 Katanning	1.0	1.26 [§]	1.26	1.26	1.37×10^{16}	1.23×10^{16}	1.52×10^{16}	4.73	Dawson <i>et al.</i> (2008)
2012/03/23 Ernabella	4.0	1.6	1.5	1.6	1.25×10^{17}	1.09×10^{17}	1.55×10^{17}	5.37	This study

*Surface-rupture length (L_{sr}) data are taken from Johnston (1994) unless otherwise specified.

[†]Moment magnitude (M_w) 5.46 ± 0.03 (derived by Johnston, 1994).

[‡]Tennant Creek involved three surface rupturing events within a 24-hour period. The surface-rupture length and moment values are the sum of these three events.

[§]Includes 3 km rupture south of Western Lake Surprise scarp (Crone *et al.*, 1992).

[§]Derived from InSAR data (Dawson *et al.*, 2008).

Figure 14: Historical stable continental earthquakes with hypocentral depths (Clark et al., 2013)

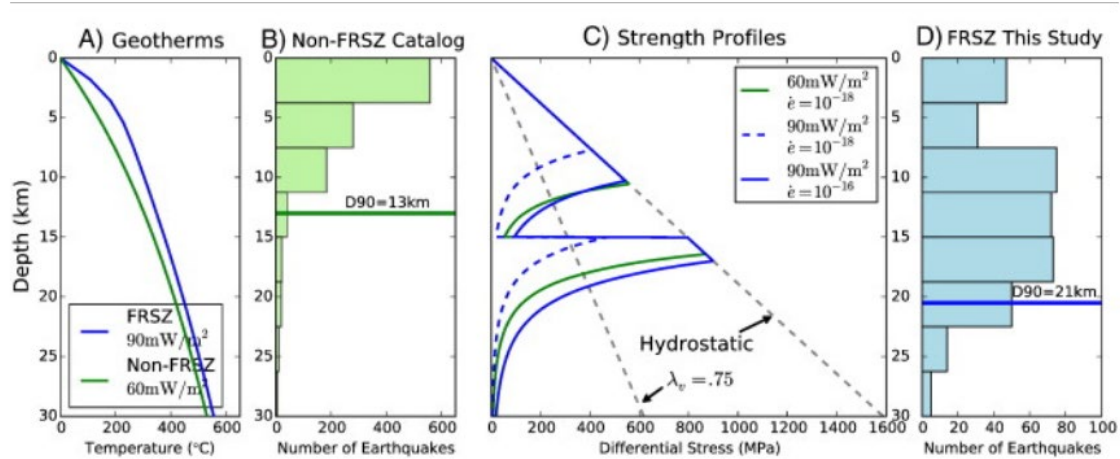


Figure 15: Depth distribution of earthquakes versus continental geotherms and strength profiles for SE Australia. (A) Continental geotherms calculated according to the model of C  lerier et al. (2005), with the hot geotherm (blue) corresponding to the high heat flow of 90 mW/m² along the axis of the Flinders Ranges, and the cooler geotherm (green) curve calculated for a heat flow of 60 mW/m² outside the Flinders Ranges; (B) depth histogram of earthquakes from the catalogue of Burbidge (2012) (excluding those for which depth was fixed arbitrarily) outside the FRSZ; (C) strength profiles using wet quartz rheology for the upper crust and wet feldspar for the lower crust, for both the 60 and 90 mW/m² geotherms of (A) and considering both low and high strain rates, 10⁻¹⁸ and 10⁻¹⁶, respectively, for the 90 mW/m² geotherm. The linear frictional failure criterion of Sibson (1974) is plotted for pore fluid pressure that is both hydrostatic and a fraction ($\lambda v = 0.75$) of lithostatic pressure. (D) Depth histogram of earthquakes recorded in this study. For (B) and (D), the line D90 indicates the bottom of the depth range which encompasses 90% of the earthquake hypocenters. Image and caption text unmodified from Balfour et al. (2015).

3.2. Seismogenic thickness

AECOM (2019) applied the Takao et al. (2013) model to reverse-faulting PFDHA analysis of the Willunga Fault and used a 0.33 weighting for the results of this model in their analysis. This model has been developed from Japanese earthquakes of all faulting types in a geologic setting (high strain rate, plate boundary, high crustal attenuation, mixed rock-sediment-soil land surface) that is not a suitable analogue for the Australian setting (low strain rate, intraplate, low crustal attenuation, bedrock-dominated land surface). AECOM (2019) assigned a 0.66 weight to the Moss and Ross (2011) model but for unknown reasons did not plot that curve on their Figure 3.

As a simple test of the Takao et al. (2013) model to the existing Australian-wide earthquake catalogue, I plot the Australian data on to Figure 3 from AECOM (2019) in Figure 16.

The Takao et al. (2013) is a highly inappropriate analogue for Australian earthquakes to-date, although the bias of Australian earthquakes to the cratonic terrains is noticed. The Australian data most-closely fit the Mixed (Petersen et al., 2011) curve but this curve is developed for strike-slip events and thus this correlation is unlikely to be meaningful in the geologic context. The Moss and Ross (2011) model is also not a good fit when compared to its successors (Moss et al., 2013). We offer an alternative curve based on empirical data from Australia in Figure 17.

My recommendation is that AECOM either use our data (Figure 17) to develop an Australia-based model that is equally weighted against Takao and Moss et al. Thrust Stiff (2013), and/or that AECOM (2019) develop a hybrid curve that combines the good fit of the Thrust Stiff (Moss et al., 2013) function to empirical data at $M < 6$ and better fit of Takao et $M > 6.2$. DrQuigs and GeoHACI could undertake an independent PFDHA analysis if this is preferable.

An appropriate weighting given existing curves could be (0.1) Takao et al, (0.1) Moss and Ross, 2011 and (0.8) Ross et al. 2013 Thrust Stiff.

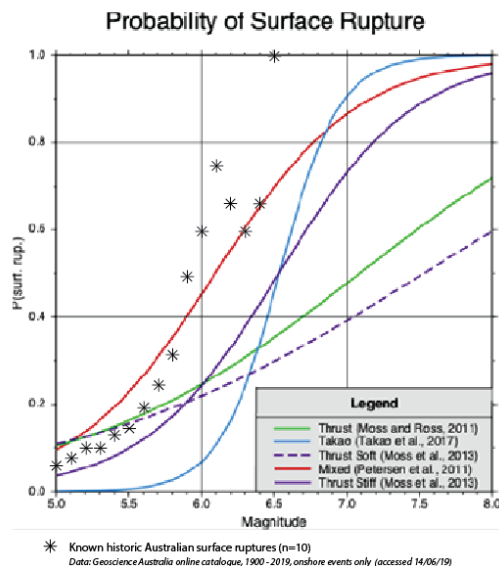


Figure 16: Australian data plotted against PFDHA curves from AECOM (2019). Takao et al. (2017) is one of the poorest fits to the Australian data, particularly at magnitude < 6 .

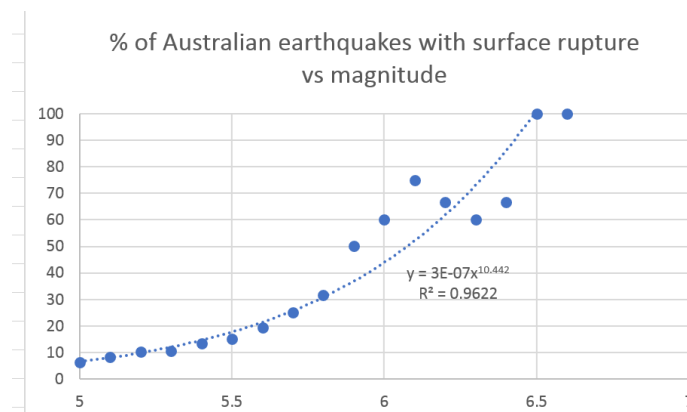


Figure 17: Power law fit to Australian data. Y axis is % of historical earthquakes out of total observed on the Australian continental landmass that have generated surface rupture; x axis is moment magnitude.

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